Build Overview:

MECHANICAL MANUFACTURING INSTRUCTIONS

CDU Water Supply Assembly - Part 1

CHANGE HISTORY

Date	BPM Rev	Change Details	Updated by
	1	New Issue	JOSEPH AMAD

Prepared By:
JOSEPH R. AMAD
Manufacturing Engineer

Checked By:
IVAN NOWELL PANIQUE
Team Lead

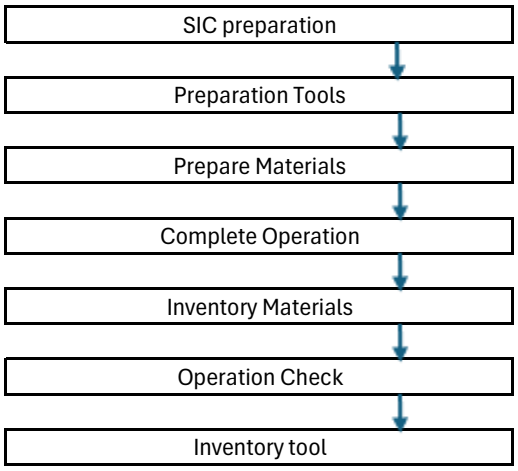
Approved By:
SHERWIN ARIOSTE
Operations Manager

SPECIAL TOOLS

1-1/2”, 41mm Crowsfoot
1.6 Nm, 5.6 Nm, 97.5 Nm Torque Wrench
1-1/2” Wrench
13mm Hex wrench
4mm Hex bit
Crimping Tools
Water Supply Orientation Fixture 654-641-40

METHOD:

Flow chart of operation steps



Size Hardware	Newton Meter
M2 x 0.4	0.2
M3 x 0.5	0.7
M4 x 0.7	1.6
M5 x 0.8	3.3
M6 x 1.0	5.6
M8 x 1.25	13.5

Torque Rules

- Torque values shown are for steel-to-steel material only.
- Torque all screws and nuts unless otherwise specified.
- Torque table is for reference. See MFE-500-01 for detailed information.

MAN: CERTIFIED OPERATOR, TECHNICIAN

NOTES

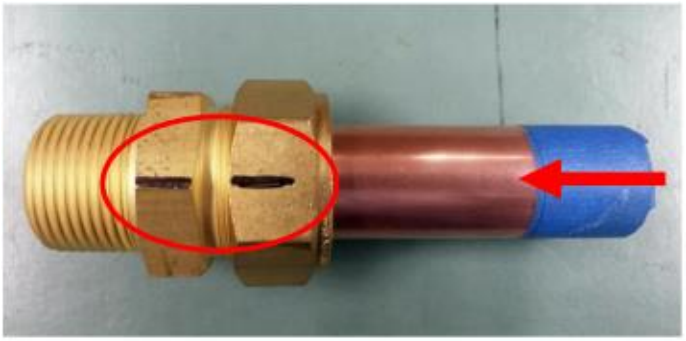
1. For all fitting requiring Loctite 565 Thread Sealant (636-666-00), apply Loctite 565 Thread Sealant to threads and spread evenly to cover a minimum of five full threads, but skip the first thread .

Wear gloves when handling the Loctite to avoid skin irritation

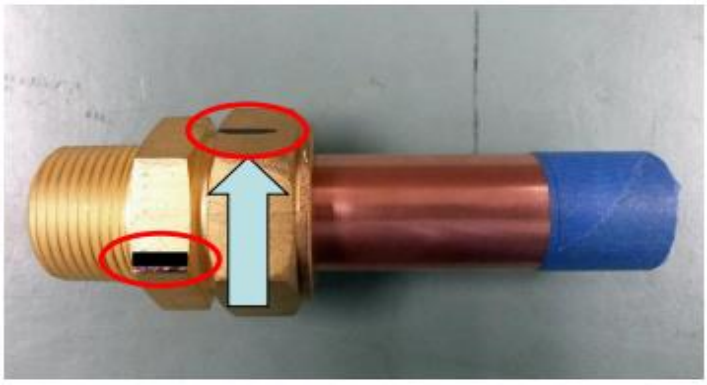
MAN: CERTIFIED OPERATOR, TECHNICIAN

NOTES (continued)

2. Process for tightening of Pipe thread and compression thread fittings
   A. Hand tighten the parts. In the case of a compression fit make sure the pipe is fully inserted into the fitting
   B. Using a “Sharpie” style black pen, make a mark in line on both components



   C. Tighten the fitting to the specification stated in the specific assembly instruction example: “Tighten 1 and ¼ turns”



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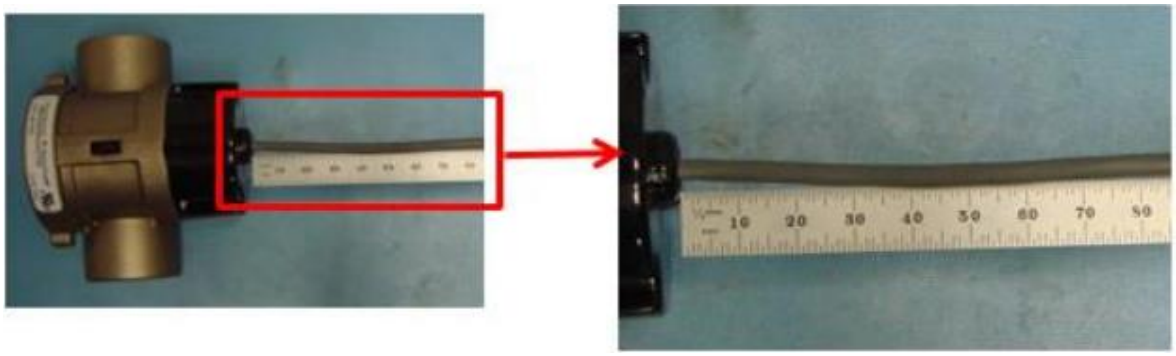
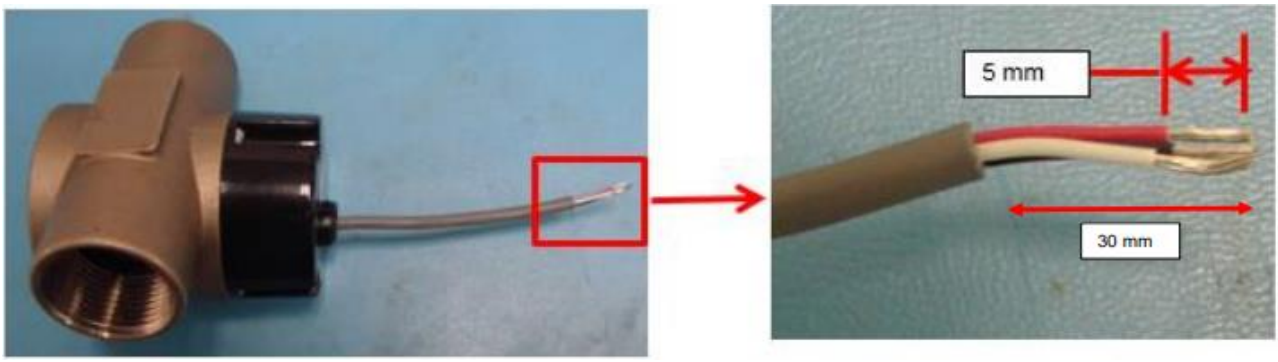
Flowmeter

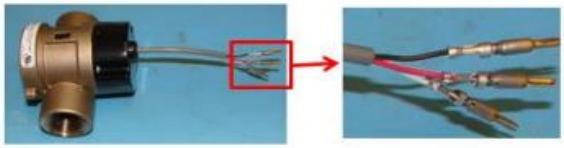
1. Cut cable from Flow meter 668-070-00 to 350mm
2. Strip outer insulation jacket to 30mm
3. Strip individual wire jackets to 5mm



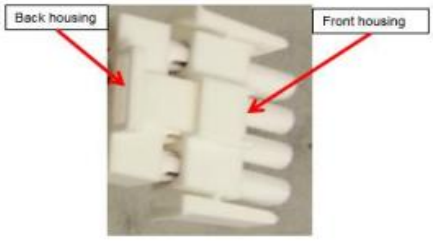


MAN: CERTIFIED OPERATOR, TECHNICIAN

5. Crimp (X3) pin connectors (634-075-01) onto wires using a standard crimping tool.

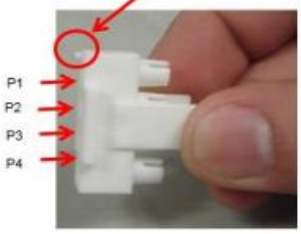


6. Obtain qty. 1 4 Pin Plug (359-494-04) kit which includes back and front housings.

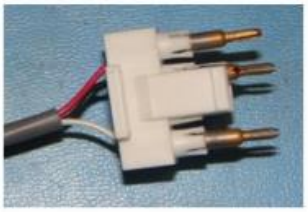


7. Take the back housing of the plug and Insert the pin connectors into the holes until they are fu seated.

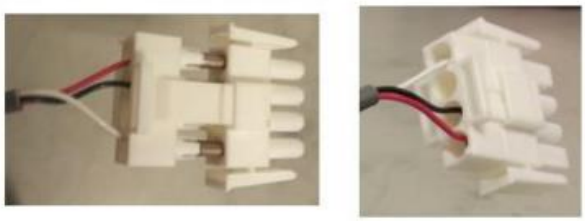
Note the orientation of the back housing. The pin hole nearest to the molded boss on the back housing designates Pin 1 hole (P1)



a. Insert Red wire into P1
b. Insert Black into P2
c. No Connection into P3
d. Insert White wire into P4



8. Install the front housing onto the back housing by snapping in place.



9. Create label and mark with location data as shown.

P20
(FT101)
FLOW METER

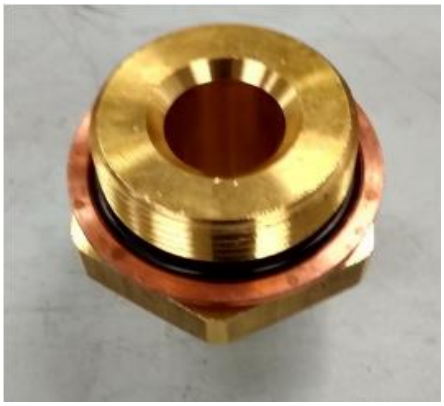
MAN: CERTIFIED OPERATOR, TECHNICIAN

Apply Molykote 111 to O-rings 470-355-07
Place O-ring over fitting 660-879-50
Place brass ring 660-680-02 over O-ring
Apply Molykote 111 to threads.
Place Flow Meter 661-198-00 in vice grip
Hand thread fitting to Flow Meter 661-198-00
    Note orientation of Valve
Use 41mm Crowsfoot wrench and torque to 97.5 Nm
    Warning: this is a high torque!
Repeat on other side of Flow Meter

660-680-02 Brass ring
660-879-50 Fitting
470-355-07 O-ring

661-198-00

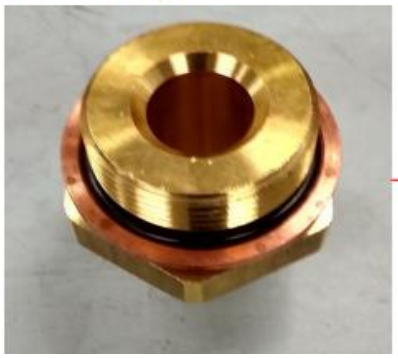
660-680-02 Brass ring
660-879-50 Fitting
470-355-07 O-ring



MAN: CERTIFIED OPERATOR, TECHNICIAN

Apply Molykote 111 to O-ring 470-355-07
Place O-ring over fitting 660-879-50
Place brass ring 660-680-02 over O-ring
Apply Molykote 111 to threads.
Place Valve 661-322-10 in vice grip
Hand thread fitting to Valve 661-322-10
Note orientation of Valve
Use 41mm Crowsfoot wrench and torque to 97.5 Nm
Warning: this is a high torque!
Repeat other side of Valve with fitting 660-879-90

660-680-02 Brass ring
660-879-50 Fitting
470-355-07 O-ring

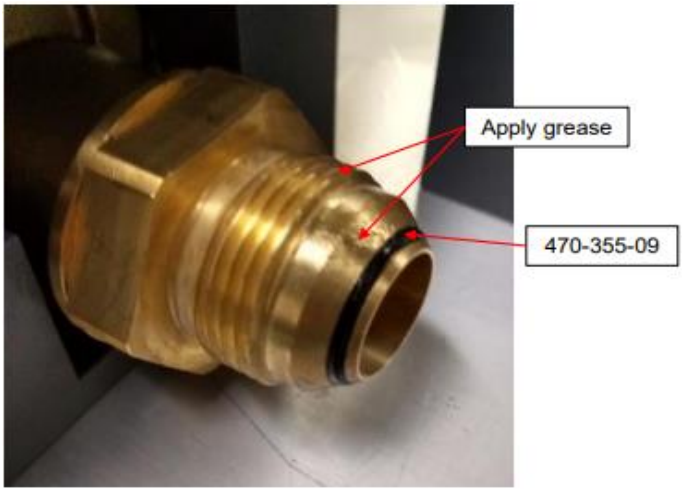


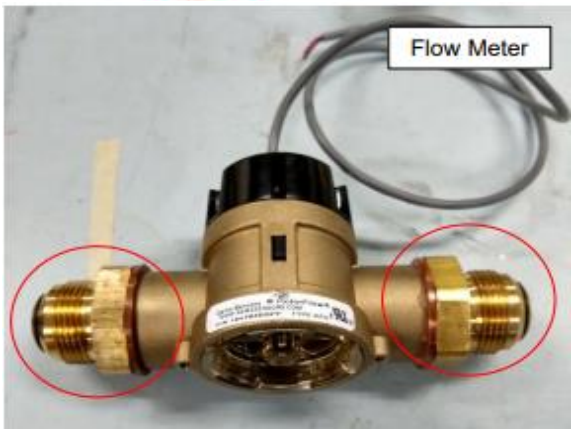
661-322-10

660-680-02 Brass Ring
660-879-90 Fitting
470-355-07 O-ring

MAN: CERTIFIED OPERATOR, TECHNICIAN

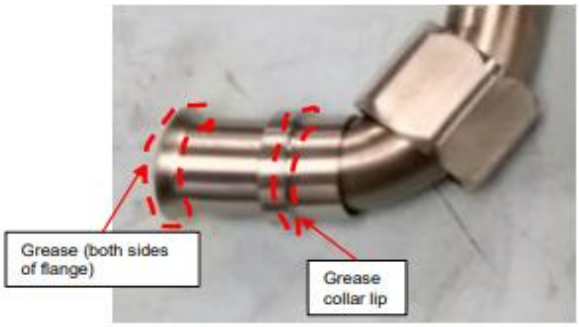
Apply Molykote 111 to O-ring 470-355-09 x3
Install O-ring to fitting 660-879-50 on flow meter and valve
Apply Molykote 111 to fitting flare and a couple threads

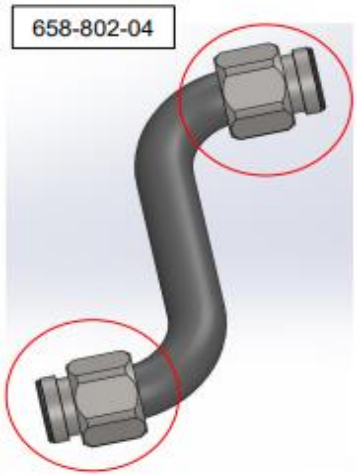




MAN: CERTIFIED OPERATOR, TECHNICIAN

Apply Molykote 111 to the flange and collar lip of Pipes 658-802-04 and 658-802-05, both ends of pipe.
Move collars up and down to ensure they are floating and not stuck.

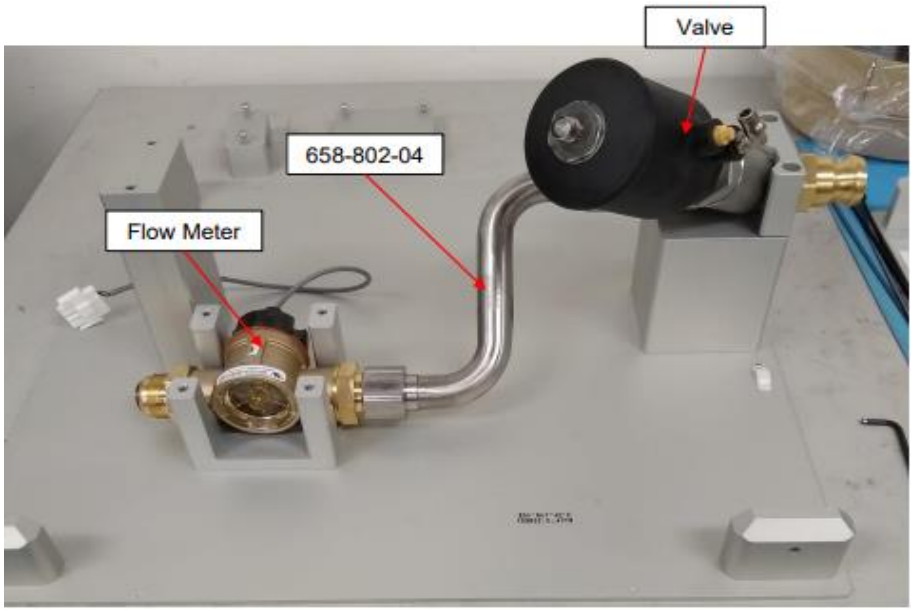


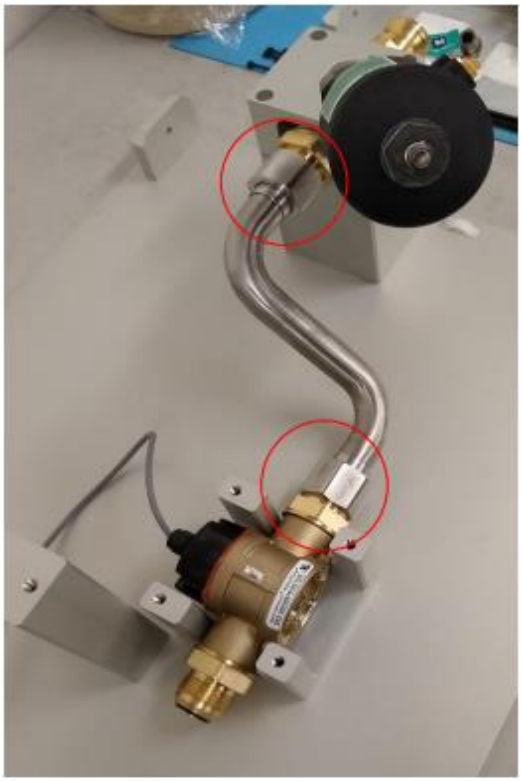


MAN: CERTIFIED OPERATOR, TECHNICIAN

Place flow meter sub-assembly into the Orientation Fixture (do not clamp)
Thread together fitting on flow meter and pipe 658-802-04. Hand tight, then back off 1/4 rotation so assemblies can still float.
Place valve sub-assembly in Orientation Fixture and hold in place (do not clamp)
Thread together pipe 658-802-04 and fitting on valve. Hand tight, then back off 1/4 rotation.
Adjust pipe, flow meter, and valve so they match the orientation shown.
Hand tight the pipe on both sides, starting with the flow meter and ending with the valve.
Clamp both flow meter and valve to Orientation Fixture. Torque screws to 1.6 Nm.
Use 1-1/2" Crowsfoot wrench and torque both to 156 Nm

Warning: this is a high torque!





Safety Tips: N/A

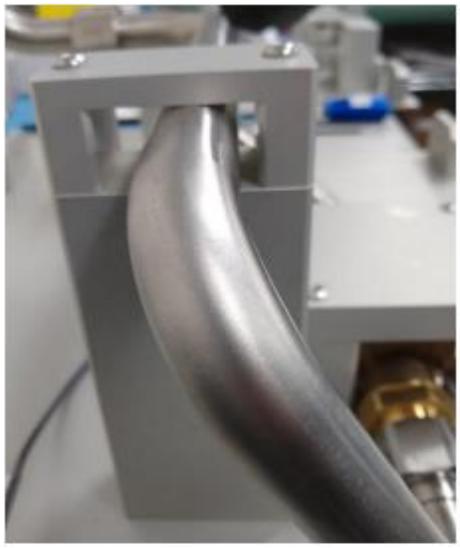
MAN: CERTIFIED OPERATOR, TECHNICIAN

Inspect unused fitting on flow meter for any damage, make sure O-ring is installed and grease Molykote 111 has been applied to fitting
Hand thread fitting and pipe 658-802-05 together to hand tight, then back off 1/4 turn so assemblies can float.
Adjust pipe so it rests on orientation fixture.

Note: Clamps on flow meter and valve may need to be loosened to adjust assembly.

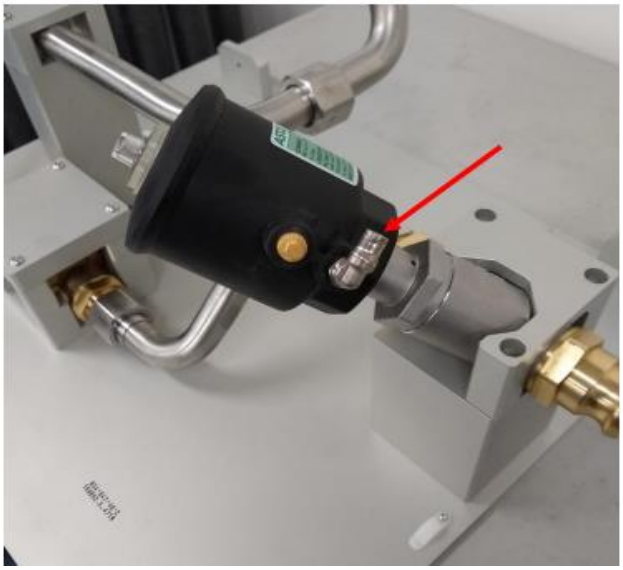
Hand tight pipe to flow meter
Clamp pipe to Orientation Fixture. Torque screws to 1.6 Nm
Use 1-1/2" Crowsfoot wrench and torque to 156 Nm

Warning: this is a high torque!



MAN: CERTIFIED OPERATOR, TECHNICIAN

Add grease to fitting 361-072-28, covering O-ring and a few threads
Install fitting to Valve
Torque to 5.6 Nm with a 13mm Wrench
Orient as shown



Part 1 complete
Perform Leak Test
If pass, continue to part 2